NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

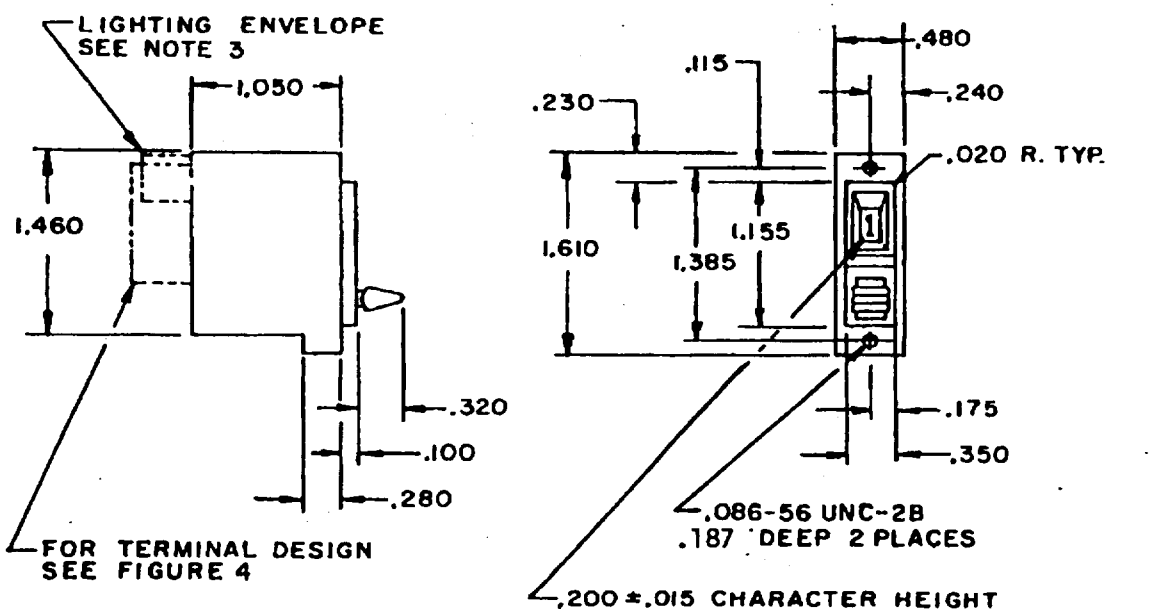
MIL-PRF-22710/24A
14 March 1977
SUPERSEDING
MIL-S-22710/24
20 June 1972

PERFORMANCE SPECIFICATION SHEET

**SWITCHES, ROTARY (PRINTED CIRCUIT)
(TOGGLE, BIDIRECTIONAL, SEALED, ILLUMINATED OR NONILLUMINATED),
STYLE SRPC24**

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switches described herein shall consist of this document and the latest issue of Specification MIL-S-22710.



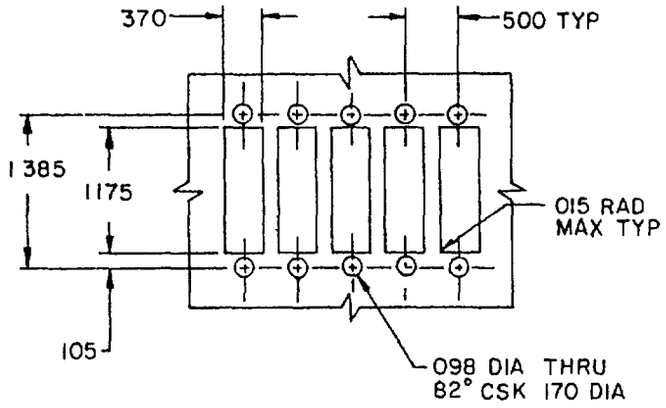
INCHES	MM	INCHES	MM	INCHES	MM
.015	.38	.18	4.75	.350	8.89
.020	.51	.200	5.08	.480	12.90
.100	2.54	.230	5.84	1.050	26.60
.086	2.18	.240	6.10	1.155	29.34
.115	2.92	.280	7.11	1.385	35.18
.175	4.45	.320	8.13	1.460	37.08
				1.610	40.89

FIGURE 1. Style SRPC24 switch and panel cutout and installation data.

(A) denotes changes

FSC 5930

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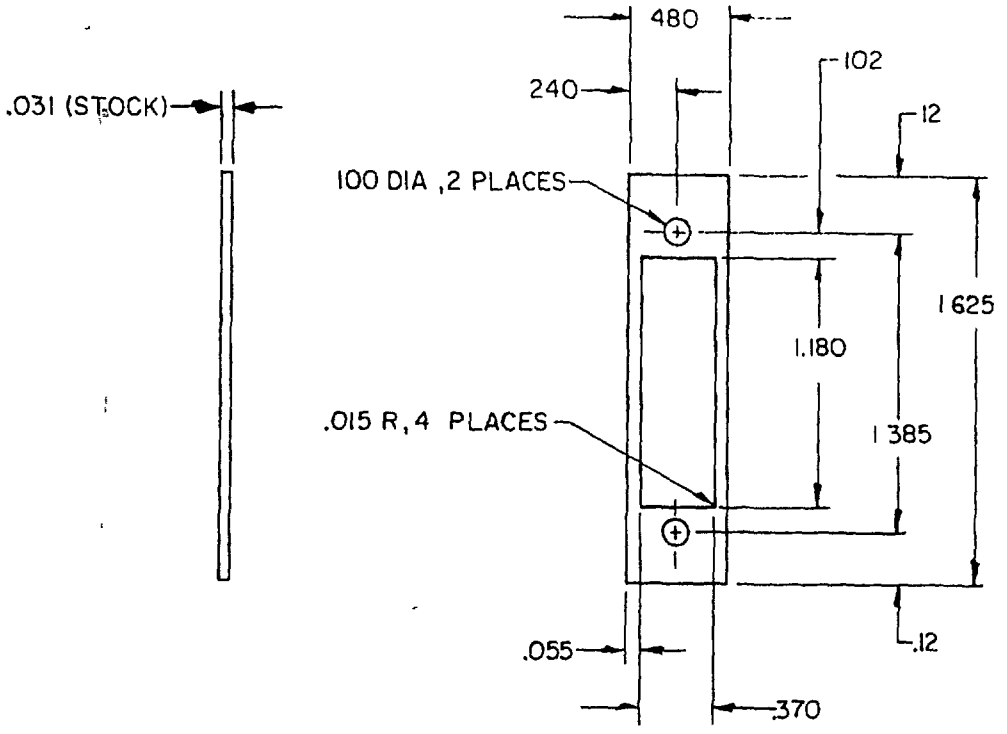
INCHES	MM
.015	.38
.098	2.49
.105	2.68
.170	4.32
.370	9.40
.500	12.70
1.175	29.85
1.385	35.18

RECOMMENDED PANEL CUTOUT

1. Dimensions are in inches, unless otherwise specified. Tolerance is ±.010 (.25MM)
2. Metric equivalents (to the nearest .01 MM) are given for general information only and are based upon 1 inch = 25.4 MM.
3. Applicable to illuminated switches only. Allowance for lighting terminals for externally replaceable lamps, when internal lighting is specified.

FIGURE 1. Style SRPC2A switch and panel cutout and installation data - Continued

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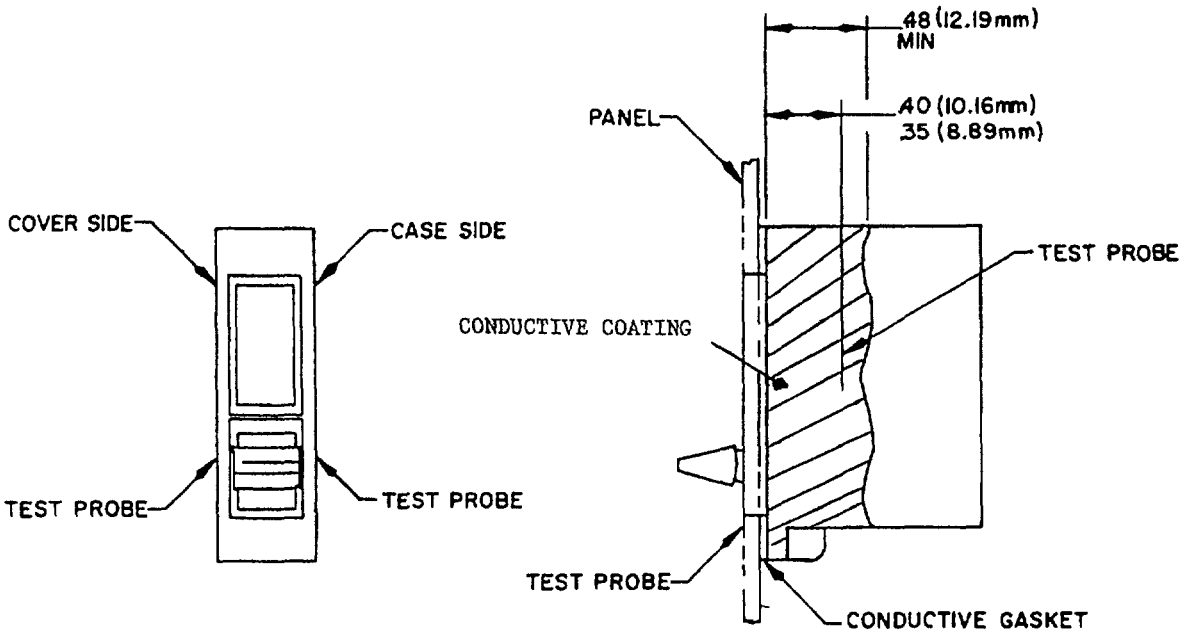


INCHES	MM	INCHES	MM
.015	.38	.240	6.10
.031	.79	370	9.40
.055	1.40	.480	12.19
.100	2.54	1 180	29.97
.102	2.59	1 385	35 18
.12	3 0	1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.02 (.5 mm) for two place decimals.
4. Panel seal material shall be neoprene rubber.

Ⓐ FIGURE 2. Panel seal gasket.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Conductive coating including front mounting surface contains 70% silver.

Ⓐ FIGURE 3. RFI shielded switch.

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REQUIREMENTS:

Dimensions and configuration. See figures 1, 2, and 3.

Construction D

(A) Angle of throw: 36°

(A) Temperature-life characteristic: B, except -55°C low temperature for all switches and +71°C for lighted switches.

Vibration grade: 3(15g, 10-2,000 Hz).

Shock type: M(100g) and H (high-impact). Methods I and II are applicable.

Moisture resistance Method 103, test condition A, MIL-STD-202, with polarization voltage.

Altitude rating C(0-70,000 feet)

Terminal design. See figure 4.

(A) TABLE I. Electrical test load.

Test or condition	Resistive	
At atmospheric pressure	0.125 Amp	28 Vdc and ac, 60 Hz
At reduced barometric pressure	0.125 Amp	28 Vdc and ac, 60 Hz
Life (low level)	10 mA	30 mV

Luminance. (For illuminated switches only.)

TABLE II. Photometric brightness.

Lamp	Switch filter	Brightness	Rating	
			Volts	Amps
MS24515-718	Clear	2.0 +1.0 foot-lamberts	5	0.115
MS24515-718	Red	1.0 +0.5 foot-lamberts	5	0 115
MS24515-718	Amber	2 0 +1.0 foot-lamberts	5	0 115

Operating force (tangential): 12 ounces minimum.
36 ounces maximum

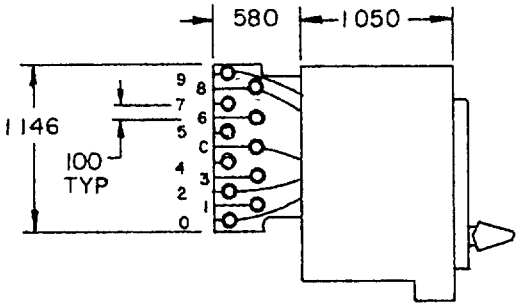
Seal: Method II

Dielectric withstanding voltage: 750 volts rms at atmospheric pressure.
250 volts rms at reduced barometric pressure, test condition C

(A) RFI shielding. Shielded switches (see table III) are intended for use in equipment required to meet the radiated (R) portions of MIL-STD-461. The conductive coating and conductive gasket shall be as shown on figure 3. When the switch is mounted facing outward to an electrically conductive panel, the resistance shall be measured between the RFI shielding on the switch and the mounting panel. The resistance shall not exceed 1 ohm.

Part number M22710/24 - (dash number from table III).

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Truth table 1/ Straight decimal.	
Dial marking	Common C Conn to
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

INCHES	MM
.100	2.54
.580	14.73
1.050	26.67
1.146	29.11

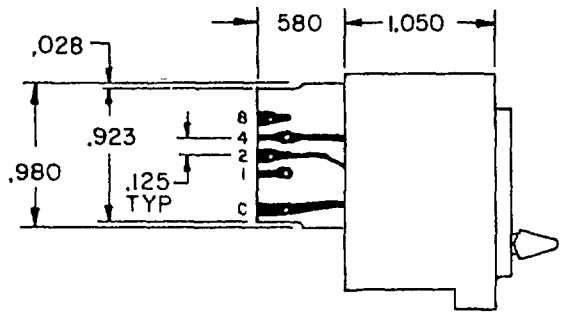
1/ Code explanation: Dial reads 0 - Common C active with terminal 0 only;
Dial reads 1 - Common C active with terminal 1 only;
Dial reads 2 - Common C active with terminal 2 only;
and so forth.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Circuit configuration 1

FIGURE 4 Module and circuit configuration.

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Truth table 1/ Binary coded decimal				
Dial marking	Common C Conn to:			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●

INCHES	MM
.028	.71
.125	3.18
.580	14.73
.923	23.44
.980	24.89
1.050	26.67

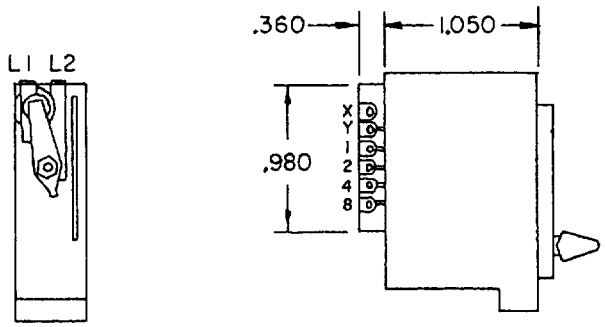
1/ Code explanation: Dial reads 0 - All terminals non-active;
Dial reads 1 - Common C active with terminal 1 only;
Dial reads 2 - Common C active with terminal 2 only;
Dial reads 3 - Common C active with terminals 1 and 2 only;
and so forth.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Circuit configuration 2

FIGURE 4. Module and circuit configuration - Continued.

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Truth table 1/ Binary coded octal with two commons				
Dial marking	Common X(●) Y(O) Conn to:			
	1	2	4	8
0	O	O	O	O
1	●	O	O	O
2	O	●	O	O
3	●	●	O	O
4	O	O	●	O
5	●	O	●	O
6	O	●	●	O
7	●	●	●	O
8	O	O	O	●
9	●	O	O	●

INCHES	MM
.360	9.14
.980	24.89
1.050	26.67

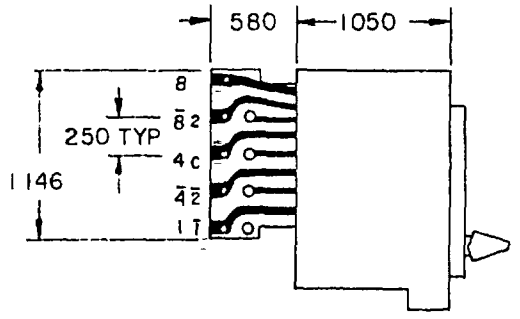
1/ Code explanation: Dial reads 0 - Common Y active with terminals 1, 2, 4 and 8 while common X is not connected to any terminals;
Dial reads 1 - Common X connected to terminal 1, while common Y is connected to terminals 2, 4, and 8,
Dial reads 2 - Common X connected to terminal 2, while common Y is connected to terminals 1, 4, and 8, and so forth

NOTES.
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Where lighting is specified, terminals shall be marked L1, L2.

Circuit configuration 3

FIGURE 4. Module and circuit configuration - Continued.

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Truth table 1/ Binary coded decimal plus compliment								
Dial marking	Common C Conn to							
	1	2	4	8	1̄	2̄	4̄	8̄
0					●	●	●	●
1	●					●	●	●
2		●			●		●	●
3	●	●					●	●
4			●		●	●		●
5	●		●			●		●
6		●	●		●			●
7	●	●	●					●
8				●	●	●	●	
9	●			●		●	●	

INCHES	MM
.250	6.35
.580	14.73
1.050	26.67
1.146	29.11

1/ Code explanation: Dial reads 0 - Common C active with terminals 1̄, 2̄, 4̄ and 8̄ only;
Dial reads 1 - Common C active with terminals 1, 2̄, 4̄ and 8̄ only;
Dial reads 2 - Common C active with terminals 2, 1̄, 4̄ and 8̄ only;
and so forth

Circuit configuration 4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 4. Module and circuit configuration - Continued.

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TABLE III. Part number and module characteristics.

Part number M22710/24-	Circuit configuration	Dial		Illumination		Special features
		Background	Marking	Color	Location	
001	1	Black Opaque	White Opaque	---	---	---
002	1	Black Opaque	White Opaque	---	---	RFI shielded
003	1	Black Opaque	White Translucent	Red	Back	---
004	1	Black Opaque	White Translucent	Red	Back	RFI shielded
005	1	Black Opaque	White Translucent	Clear	Back	--
006	1	Black Opaque	White Translucent	Clear	Back	RFI shielded
007	2	Black Opaque	White Opaque	---	---	---
008	2	Black Opaque	White Opaque	---	---	RFI shielded
009	2	Black Opaque	White Translucent	Red	Back	---
010	2	Black Opaque	White Translucent	Red	Back	RFI shielded
011	2	Black Opaque	White Translucent	Clear	Back	---
012	2	Black Opaque	White Translucent	Clear	Back	RFI shielded
013	3	Black Opaque	White Opaque	---	---	---
014	3	Black Opaque	White Opaque	---	---	RFI shielded
015	3	Black Opaque	White Translucent	Red	Back	---
016	3	Black Opaque	White Translucent	Red	Back	RFI shielded
017	3	Black Opaque	White Translucent	Clear	Back	---
018	3	Black Opaque	White Translucent	Clear	Back	RFI shielded
019	3	Black Opaque	White Translucent	Amber	Back	---
020	3	Black Opaque	White Translucent	Amber	Back	RFI shielded
021	4	Black Opaque	White Opaque	---	---	---
022	4	Black Opaque	White Opaque	---	---	RFI shielded
023	4	Black Opaque	White Translucent	Red	Back	---
024	4	Black Opaque	White Translucent	Red	Back	RFI shielded
025	4	Black Opaque	White Translucent	Clear	Back	---
026	4	Black Opaque	White Translucent	Clear	Back	RFI shielded

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Air Force - 85

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Navy -
Air Force - 17, 99
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Air Force - 11

Preparing activity:
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Agent:
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